

Shadow · Canary · A/B

Your new model scores 3% higher offline. You ship it. Conversion drops 2%. Revenue drops 5%. What happened? Only a production A/B would have told you.

When did you last ship a model CHANGE without an A/B test?

WHAT'S INSIDE

- 01 Three deploy patterns
- 02 Sample size — the formula
- 03 Success + guardrail metrics
- 04 Peeking + multiple testing
- 05 Analogy

WHY THIS LESSON MATTERS

Offline winner. Production loser.

NUMBER 1

3%

From real-world deployments.

NUMBER 2

2%

From real-world deployments.

NUMBER 3

5%

From real-world deployments.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Shadow vs canary vs A/B deploys.

2

Compute sample size.

3

Pick success + guardrail metrics.

4

Avoid peeking + multiple-testing errors.

5

Run an A/B on a model change.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Three deploy patterns

Progression: Shadow → Canary →
A/B → 100%.

02

Sample size — the formula

Compute BEFORE you start

03

Success + guardrail metrics

One success. 2-3 guardrails.

04

Peeking + multiple testing

Silent bias amplifiers

01

PART 1 OF 4

Three deploy patterns

Progression: Shadow → Canary → A/B → 100%.

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DEFINITION

Three deploy patterns

Progression: Shadow → Canary → A/B → 100%.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Sample size — the formula

Detect 1% lift on 10% baseline; 80% power; $\alpha=0.05$.

$n \approx 13,900$ per variant.

Use `scipy.stats.norm.ppf`.

Commit to that n . Never peek.

COMMITMENT

Set n . Stop.

Peeking is silently-wrong.

KEY POINTS

- Detect 1% lift on 10% baseline; 80% power; $\alpha=0.05$
- $n \approx 13,900$ per variant
- Use `scipy.stats.norm.ppf`
- Commit to that n . Never peek

CONCEPT 3 OF 4

Success + guardrail metrics

Success: the ONE number you want to move (conversion rate).

Guardrails: must-not-degrade (latency, error rate, retention).

North-star: company KPI — reference only.

Never 10 metrics. Always find "a win".

RULE

1 + 2-3

1 success + 2-3 guardrails.

DEFINITION

Success + guardrail metrics

One success. 2-3 guardrails.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Peeking + multiple testing

Peeking inflates false-positive rate.

Fix: sample-size pre-commit; check at end only.

Or use sequential testing (harder).

Multiple metrics → Bonferroni correction.

DISCIPLINE

No peeking

Rigour over convenience.

KEY POINTS

- Peeking inflates false-positive rate
- Fix: sample-size pre-commit; check at end only
- Or use sequential testing (harder)
- Multiple metrics → Bonferroni correction

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where will you apply this in your next project?

Q2

What was the single biggest 'aha' from this lesson?

Q3

Which habit will you start using this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Why pre-commit n?

A

Speed

B

Resist peeking

C

Fun

D

Rule

Why pre-commit n ?

CORRECT ANSWER



Resist peeking

WHY

Peeking inflates false positives

Success metrics per test:



B



D



As many as you want



(no option)

Success metrics per test:



CORRECT ANSWER

As many as you want

WHY

One success metric. 2-3 guardrails

Offline winner → production loser because:

A

Never happens

B

Latency + selection bias

C

Random

D

Only in LLMs

Offline winner → production loser because:

CORRECT ANSWER



Latency + selection bias

WHY

Offline misses deployment effects

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Progression: Shadow → Canary → A/B → 100%.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

- 1 Three deploy patterns
- 2 Sample size — the formula
- 3 Success + guardrail metrics
- 4 Peeking + multiple testing

Calculate. Commit. Never peek.

"Rigour over convenience."

DELIVERABLES

- NEXT
- Lesson 7 — Cost optimisation.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.